

PRIYANSHU KUMAR

Aspiring FPGA Design Engineer | Digital Design & Embedded Systems
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SUMMARY

Computer & Communication Engineering student with a strong foundation in digital design fundamentals (combinational & sequential logic, FSMs, timing, CDC) and hands-on experience in Verilog/VHDL, FPGA prototyping on Xilinx Vivado, and circuit simulation. Complements hardware skills with full-stack web development (React, Node.js) and IoT/embedded projects, bringing an end-to-end perspective on hardware–software integration.

TECHNICAL SKILLS

Digital Design: Combinational & Sequential Logic, FSMs, Timing Analysis, Clock Domain Crossing (CDC) concepts, VLSI

HDLs: Verilog, VHDL — synthesizable coding & simulation

FPGA Tools: Xilinx Vivado (synthesis, place-and-route, timing analysis), waveform analysis / testbenches

Embedded & Hardware: Embedded Systems & Programming, Circuit Design, IoT, Signal Processing, LTSpice

Programming: C, C++, Java, JavaScript

Full-Stack Development: React, Node.js, Express, MongoDB, HTML, CSS

Platforms & Tools: MATLAB, Scilab, Git/GitHub, RedHat Linux, MS Excel/Word

EDUCATION

B.Tech in Computer and Communication Engineering

Expected May 2027

JK Lakshmipat University, Jaipur, Rajasthan

EXPERIENCE

VHDL / FPGA Design Intern | IEST Shibpur, Kolkata

May 2025 – Aug 2025

- Designed and implemented digital logic modules in VHDL, translating functional specifications into synthesizable RTL for FPGA targets.
- Practiced simulation and waveform-based verification of digital circuits, gaining exposure to FPGA design flow from RTL to implementation.
- Contributed to embedded system development, reinforcing hardware–software integration fundamentals.

Cadence Software Hands-on Training | Punjab Engineering College (PEC)

- Built proficiency in Cadence for circuit simulation and analysis, including schematic capture and functional verification.
- Completed NAND and NOR gate circuit analysis projects, strengthening fundamentals in combinational logic design.

PROJECTS

Automated Vehicle Lane-Changing System — FPGA · VHDL/Verilog

- Designed a real-time FPGA-based control system for automated lane changing using VHDL/Verilog, including FSM-driven decision logic.

Arithmetic Logic Unit (ALU) — FPGA · Digital Design

- Implemented and verified a multi-function ALU on FPGA through RTL design and simulation-based testing.

Traffic Light Controller — FPGA · FSM Design

- Developed an FSM-based controller for efficient, rule-driven traffic signal management and timing control.

UART Communication Module — Digital Design

- Implemented a UART transmitter/receiver for serial data communication between digital systems, verified via testbench simulation.

FIFO Buffer Design — Digital Design

- Designed and verified a synchronous FIFO buffer for reliable data buffering between digital modules with different timing.

MeetCut — Full-Stack Web App

- Built a meeting-scheduling platform with time-slot booking and calendar management using a React frontend and a Node.js/Express API.

UrbanNest — Full-Stack Web App

- Developed a real estate/property-listing platform with search, filtering, and a management dashboard.

ACCOMPLISHMENTS

- Completed multiple hands-on projects spanning signal processing, digital electronics, and FPGA-based design.
- Gained practical experience in Cadence for NAND/NOR gate circuit analysis.
- Participated in the 5G Use Case Lab workshop, building awareness of telecommunications systems.

HOBBIES

Playing Cricket • Communicating with People

REFERENCES

Dr. Devika Kataria — devikakataria@jklu.edu.in

Dr. Gaurav Mani Khanal — Gauravmanikhanal@jklu.edu.in